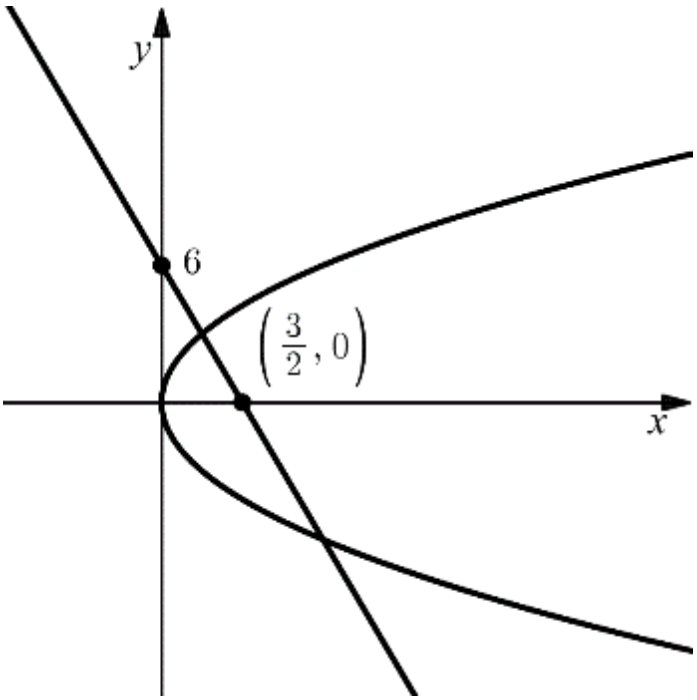
1.26 Conics: Problem Solving

Worked Solutions

1 a)



1 b)

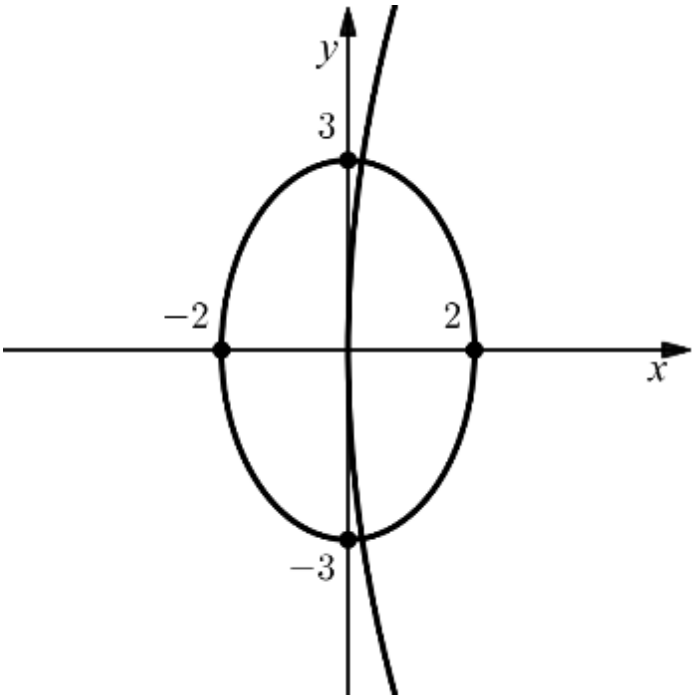
$$\begin{aligned}(6-4x)^2 &= 12x \\ 36-48x+16x^2 &= 12x \\ 16x^2-60x+36 &= 0 \\ \therefore x &= \frac{3}{4} \text{ or } x = 3\end{aligned}$$

So $(\frac{3}{4}, 3)$ and $(3, -6)$

1 c)

$\sqrt{9^2 + (\frac{9}{4})^2} = \frac{9\sqrt{17}}{4}$

2 a)



2 b)

$$\frac{x^2}{4} + \frac{40x}{9} = 1$$
$$9x^2 + 160x = 36$$
$$9x^2 + 160x - 36 = 0$$
$$(9x - 2)(x + 18) = 0$$
$$x = \frac{2}{9} \text{ or } x = -18$$

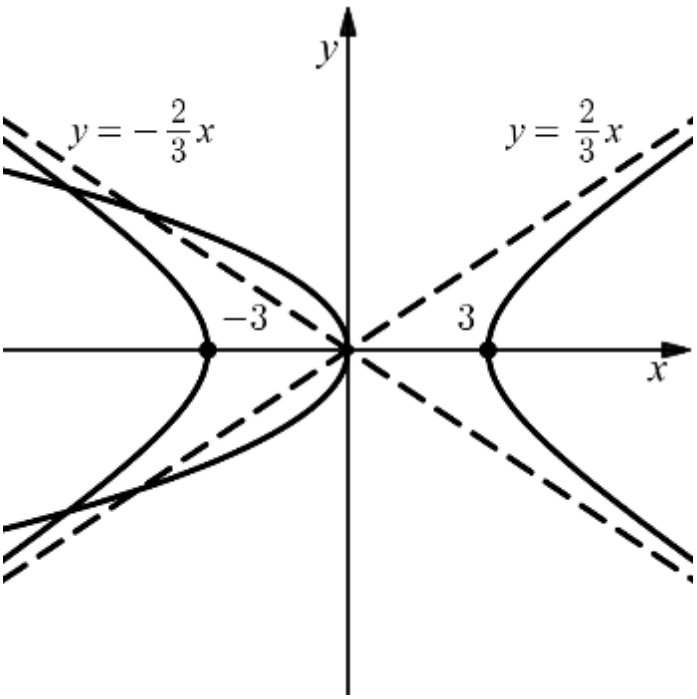
So since x is positive

$$x = \frac{2}{9}$$

$y^2 = 40 \times \frac{2}{9} \Rightarrow y = \pm\frac{4\sqrt{5}}{3}$

$$(\tfrac{2}{9}, -\tfrac{4\sqrt{5}}{3}) \text{ and } (\tfrac{2}{9}, \tfrac{4\sqrt{5}}{3})$$

3 a)



3 b)

$$\frac{x^2}{9} - \frac{(-2x)}{4} = 1$$
$$4x^2 + 18x = 36$$
$$4x^2 + 18x - 36 = 0$$
$$2x^2 + 9x - 18 = 0$$
$$(x + 6)(2x - 3) = 0$$
$$x = -6 \text{ or } x = \frac{3}{2}$$

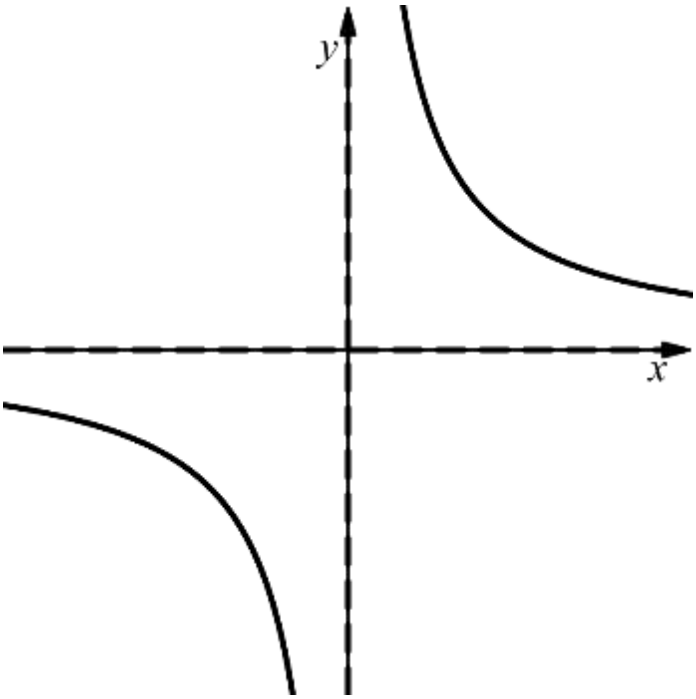
So since x is negative

$$x = -6$$

$y^2 = -2 \times -6 \Rightarrow y = \pm 2\sqrt{3}$

$$(-6, -2\sqrt{3}) \text{ and } (-6, 2\sqrt{3})$$

4 a)



4 b)

$$\begin{aligned} x(-kx + 8\sqrt{2}) &= 16 \\ -kx^2 + 8\sqrt{2}x - 16 &= 0 \\ kx^2 - 8\sqrt{2}x + 16 &= 0 \\ \therefore x &= \frac{8\sqrt{2} \pm \sqrt{128 - 64k}}{2k} = \frac{4\sqrt{2} - 4\sqrt{2 - k}}{k} \end{aligned}$$

4 c)

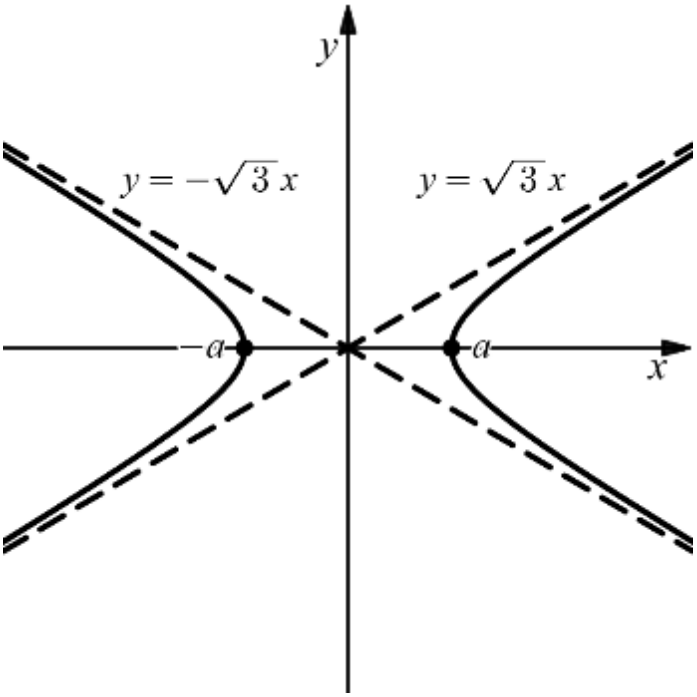
$2 - k = 0$ so $k = 2$

5 a)

$y = \pm\frac{\sqrt{3a^2}}{a}x = \pm\sqrt{3}a$

$y = \sqrt{3}a$ and $y = -\sqrt{3}a$

5 b)

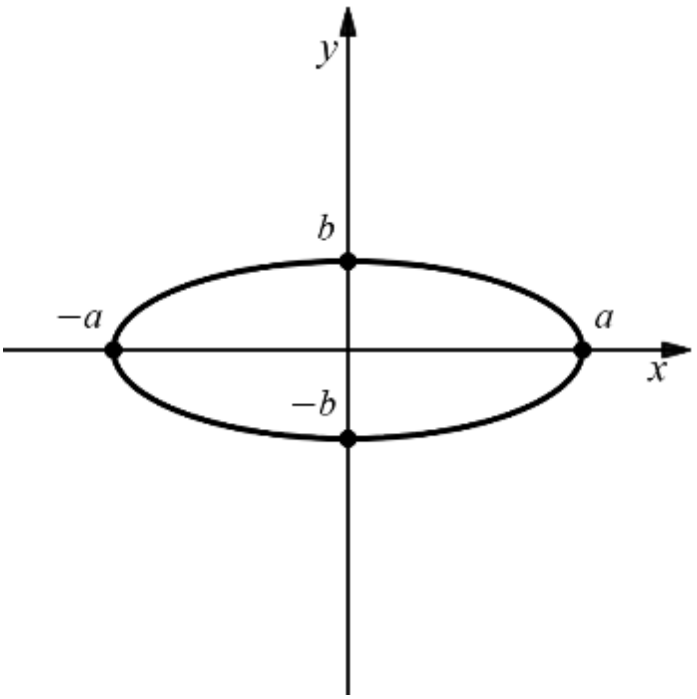


5 c)

$\frac{y^2}{3a^2} = \frac{x^2}{a^2} - 1 \Rightarrow y^2 = 3x^2 - 3a^2$

$$\begin{aligned}\pi \int_a^{2a} 3x^2 - 3a^2 \,\mathrm{d}x &= \pi[x^3 - 3a^2x]_a^{2a} \\ &= \pi[(8a^3 - 6a^3) - (a^3 - 3a^3)] \\ &= 4a^3\pi \text{ units}^3\end{aligned}$$

6 a)



6 b)

$\frac{x^2}{a^2} = 1 - \frac{y^2}{b^2} \Rightarrow x^2 = a^2 - \frac{a^2}{b^2}y^2$

$$\begin{aligned}\pi \int_0^b a^2 - \frac{a^2}{b^2}y^2 \, \mathrm{d}y &= \pi[a^2 y - \frac{a^2}{3b^2}y^3]_0^b \\ &= \pi[(a^2 b - \tfrac{1}{3}a^2 b) - (0)] \\ &= \frac{2}{3}a^2 b\pi \text{ units}^3\end{aligned}$$

7)

$$\begin{aligned}m^2x^2 &= 12x - 12k \\ m^2x^2 - 12x + 12k &= 0\end{aligned}$$

For it to be a tangent, $b^2 - 4ac = 0$

$$\begin{aligned}(-12)^2 - 4m^2 \times 12 &= 0 \\ 144 - 48m^2k &= 0 \\ 48m^2k &= 144 \\ m^2 &= \frac{3}{k} \\ \therefore m &= \pm\sqrt{\frac{3}{k}} \text{ as req.}\end{aligned}$$

8)

$$\begin{aligned}\frac{x^2}{4} + \frac{(mx+5)^2}{16} &= 1 \\ 4x^2 + (mx+5)^2 &= 16 \\ 4x^2 + m^2x^2 + 10mx + 25 &= 16 \\ (4+m^2)x^2 + 10mx + 9 &= 0\end{aligned}$$

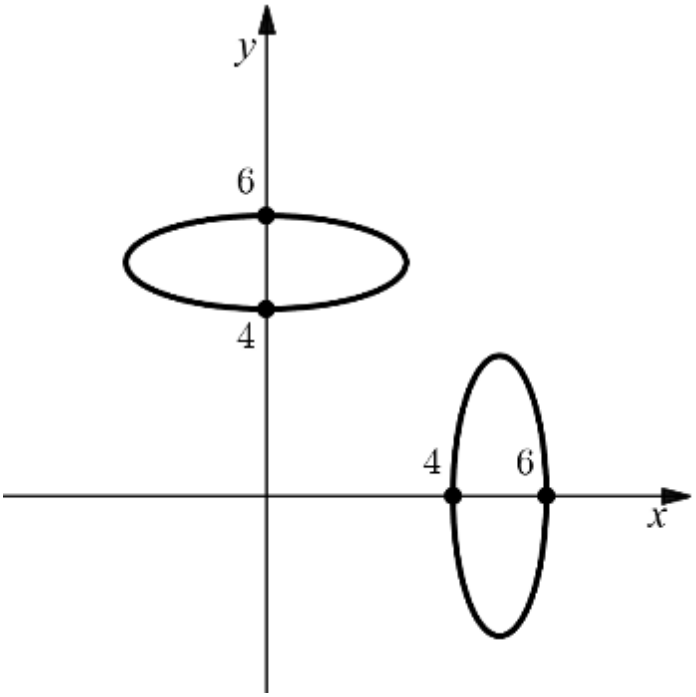
For it to be a tangent, $b^2 - 4ac = 0$

$$\begin{aligned}(10m)^2 - 4(4+m^2)(9) &= 0 \\ 100m^2 - 144 - 36m^2 &= 0 \\ 64m^2 - 144 &= 0 \\ m^2 &= \frac{9}{4} \\ m &= -\frac{3}{2} \text{ or } m = \frac{3}{2}\end{aligned}$$

9 a)

E_1: when $y = 0$, $(x-5)^2 = 1$ so $x = 4$ or $x = 6$

E_2: when $x = 0$, $(y-5)^2 = 1$ so $y = 4$ or $y = 6$



9 b)

Tangent line for E_1:

$$\begin{aligned}(x-5)^2+\frac{(k-x)^2}{9}&=1\\ 9(x-5)^2+(k-x)^2&=9\\ 9x^2-90x+225+k^2-2kx+x^2&=9\\ 10x^2-(90+2k)x+(216+k^2)&=0\end{aligned}$$

For it to be a tangent, $b^2-4ac=0$

$$\begin{aligned}(-(90+2k))^2-4\times 10(216+k^2)&=0\\ 4k^2+360k+8100-40k^2-8640&=0\\ -36k^2+360k-540&=0\\ k^2-10k+15&=0\\ k=5+\sqrt{10}\text{ or }k&=5-\sqrt{10}\end{aligned}$$

9 c)

Tangent line for E_2:

$$\frac{(k-y)^2}{9}+(y-5)^2=1$$

Which is the same as seen in part b) to give $10y^2-(90+2k)y+(216+k^2)=0$

$$\therefore k=5+\sqrt{10}\text{ or }k=5-\sqrt{10}$$

10)

Given the general point $P(x, y)$

Distance to point: $\sqrt{x^2 + (y+4)^2}$

Distance to line: $|y - 2|$

$$\begin{aligned}
\sqrt{x^2 + (y+4)^2} &= |y-2| \\
x^2 + (y+4)^2 &= (y-2)^2 \\
x^2 + y^2 + 8y + 16 &= y^2 - 4y + 4 \\
x^2 &= -12y - 12 \\
x^2 &= -12(y+1)
\end{aligned}$$

11)

Given the general point $P(x, y)$

Distance to point: $\sqrt{x^2 + (y-9)^2}$

Distance to line: $|x - 1|$

$$\begin{aligned}
\sqrt{(x-9)^2 + y^2} &= 3|x-1| \\
(x-9)^2 + y^2 &= 9(x-1)^2 \\
x^2 - 18x + 81 + y^2 &= 9x^2 - 18x + 9 \\
y^2 &= 8x^2 - 72 \\
8x^2 - y^2 &= 72 \\
\frac{x^2}{9} - \frac{y^2}{72} &= 1
\end{aligned}$$